# Year 13 Further Maths - Applications of matrices

Similar questions - Invariant lines, shears and combined transformations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

**1** The matrix $\begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix}$ represents a transformation $T$.

**(a)** The line $y = kx$ is invariant under $T$. Determine the value of $k$. **[4]**

**(b)** Determine whether $y = kx$ is a line of invariant points. **[1]**

**(c)** Find the image of the point $(2, -2)$ under $T$. **[1]**

**2**

**(a)** $\mathbf{A} = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$. Describe fully the transformation represented by **A**, and show that the area of any shape is unchanged by it. **[3]**

$\mathbf{B} = \begin{pmatrix} 7 & 3 \\ 10 & 5 \end{pmatrix}$ is the result of applying three transformations in this order: a shear that leaves the $y$-axis invariant and maps $(1,1)$ to $(1,3)$; then the transformation represented by **A**; then a stretch of scale factor $p$ that leaves the $x$-axis invariant.

(b) Determine $p$. **[4]**

**Question 1**

(a) $\begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ k \end{pmatrix} = \begin{pmatrix} 4+k \\ -1+2k \end{pmatrix}$. Invariant needs $-1+2k = k(4+k)$, so $k^2+2k+1=0$ and $k=-1$

(b) $T(1,-1) = (3,-3) \neq (1,-1)$, so the points are not fixed: $y=-x$ is not a line of invariant points

(c) $\begin{pmatrix} 4 & 1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 2 \\ -2 \end{pmatrix} = \begin{pmatrix} 6 \\ -6 \end{pmatrix}$

**Question 2**

(a) A shear with the $x$-axis invariant (a line of invariant points), mapping $(x,y)$ to $(x+3y,y)$, e.g. $(0,1) \mapsto (3,1)$. $\det \mathbf{A} = 1$ so areas are unchanged

(b) The first shear is $\begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}$ with $(1,1) \mapsto (1,1+s) = (1,3)$, so $s=2$. The stretch is $\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}$. The combined matrix is $\begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix}\begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 3 \\ 2p & p \end{pmatrix}$. Comparing with $\mathbf{B}$: $p=5$